SageMath: A Deep Dive into Graphs

In the past few articles in this series we discussed our first mathematical topic of great interest: graph theory. Specifically, we explored undirected simple graphs, both theory and practical coding. By the end of the third article in this series (published in the July 2024 issue of *OSFY*), we began focusing our attention on another important class of graphs: undirected multigraphs. We will resume our discussion from where we left off. Please refer to the program '*mgraph7.sagews*' and the multigraph *Mgraph_7* (Figure 5) generated by this program, given in the previous article in this series, before proceeding further with this article for a clear understanding. Recall that we also had a detailed discussion about the differences between undirected simple graphs and multigraphs in the previous article. Furthermore, any theoretical concept not elaborately explained in this article is discussed in detail in one of the previous articles.

The plan is to test every method we tested on the undirected simple graphs we generated using SageMath in the previous articles on *Mgraph_7*. By doing this, we will know which method does not work for undirected multigraphs and how they behave if they do work. First, let us print the adjacency matrix of the undirected multigraph *Mgraph_7*. To do that, add the following line of code at the end of the program called '*mgraph7.sagews*'.

```
print(MG.adjacency_matrix( ))
```

As before, we use the online tool called CoCalc to execute the modified program. Upon execution of the modified program, the following additional output will be displayed on the screen. But how is it different from the adjacency matrix of an undirected simple graph?

```
[1 2 0 2]
[2 1 2 0]
[0 2 1 2]
[2 0 2 1]
```

First, the adjacency matrix of an undirected simple graph only contained 0s or 1s as entries, indicating that there could be at most one edge between any pair of vertices. However, in the adjacency matrix shown above, we also have 2s, indicating that there are 2 parallel edges between certain pairs of vertices in the undirected multigraph *Mgraph_7*. Notice that there can be any number of edges between a pair of vertices in an undirected multigraph, and hence the corresponding adjacency matrix can contain any number greater than or equal to 0. Second, the diagonal elements of the adjacency matrix of an undirected simple graph contain only 0s, indicating that such graphs do not contain any loops.